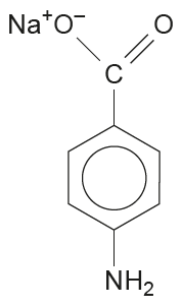


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			$E_1 \times \lambda_1 = E_2 \times \lambda_2$ (1) $\lambda_2 = \frac{E_1 \times \lambda_1}{E_2} = 267 \text{ kJ mol}^{-1}$ (1) alternative method constant = $E \times \lambda = 1.2 \times 10^5$ (1) $E = \frac{1.2 \times 10^5}{450} = 267 \text{ kJ mol}^{-1}$ (1)		1	1	2	1	
	(c)	(i)	I	 accept if Na ⁺ not included			1	1		
			II	the attacking reagent / OH ⁻ is a nucleophile(1) lone pair on N becomes part of delocalised system / C—N bond is stronger when directly attached to ring (1)		2		2		